

CPPDescent

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Chapter 1

CPP-Descent

Part of the “Software Development for Computing Systems” course of the Department of Informatics and Telecommunications, UoA for the first semester of the academic year 2023-2024.

A library that implements the KNN-Graph creation and search algorithms described in [1], [2].

1.1 Dependencies

For local development, you will need Make and CMake. To install them on an apt-based linux distro, run the following commands:

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install make
$ sudo apt install cmake
```

1.1.1 Minor dependencies

- The project uses Google Test for its testing needs. Gtest is used as a git submodule of this repo, that is downloaded by CMake the first time you clone. That means that nothing extra needs to be installed to run the tests.
- If you would like to generate the documentation, you will need doxygen installed.

1.2 Code Structure

In the [wiki](#), you can find documentation generated by Doxygen. It is also available in [pdf form](#).

1.2.1 Code Coverage

This project uses LCOV for its code coverage needs. The results are uploaded upon commit [here](#).

1.3 Contributors

- [Konstantinos Chousos](#) (1115202000215)
- [Anastasios-Fedon Seitanidis](#) (1115202000179)

1.4 Bibliography

[1] W. Dong, C. Moses, and K. Li, “Efficient k-nearest neighbor graph construction for generic similarity measures,” in *Proceedings of the 20th international conference on World wide web*, in WWW '11. New York, NY, USA: Association for Computing Machinery, Mar. 2011, pp. 577–586. doi: [10.1145/1963405.1963487](#).

[2] “How PyNNDescent works — pynndescent 0.5.0 documentation.” Accessed: Oct. 10, 2023. [Online]. Available: https://pynndescent.readthedocs.io/en/latest/how_pynndescent_works.html

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 3

File Index

3.1 File List

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Chapter 4

Class Documentation

4.1 cppdescent Class Reference

A sample function for the cppdescent class, used to test the correctness of the CMake files.

```
#include <cppdescent.hpp>
```

Public Member Functions

- int [hello](#) (void)
Prints "Hello World".

4.1.1 Detailed Description

A sample function for the cppdescent class, used to test the correctness of the CMake files.

4.1.2 Member Function Documentation

4.1.2.1 hello()

```
int cppdescent::hello (  
    void )
```

Prints "Hello World".

Returns

int exit code

The documentation for this class was generated from the following files:

- [cppdescent.hpp](#)
- [cppdescent.cpp](#)

4.2 List Class Reference

ADT [List](#).

```
#include <ADTList.hpp>
```

Public Member Functions

- [List](#) (DestroyFunc value=nullptr)
Construct a new [List](#) object.
- [~List](#) ()
Destroy the [List](#) object.
- void [insertNext](#) ([ListNode](#) *node, Pointer value)
Adds a node after the given node.
- void [removeNext](#) ([ListNode](#) *node)
Removes the node after the given node.
- Pointer [find](#) (Pointer value, CompareFunc compare)
Finds the first value equal to the value parameter.
- DestroyFunc [setDestroyValue](#) (DestroyFunc value)
Set the Destroy Value object to a new value.
- int [getSize](#) ()
Get the size of the [List](#).
- int [increaseSize](#) ()
Increase the list size.
- int [decreaseSize](#) ()
Decrease the list size.
- [ListNode](#) * [getHead](#) ()
Getter for the head of the [List](#).
- [ListNode](#) * [getTail](#) ()
Getter for the tail of the [List](#).
- [ListNode](#) * [next](#) ([ListNode](#) *node)
Returns the node after the given one.
- Pointer [nodeValue](#) ([ListNode](#) *node)
Returns the value of the node.
- [ListNode](#) * [findNode](#) (Pointer value, CompareFunc compare)
Finds the first node that has value equal to the value parameter.

4.2.1 Detailed Description

ADT [List](#).

An Abstract Data Type [List](#) with linear access to data, and addition and removal to/from arbitrary positions.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 List()

```
List::List (
    DestroyFunc value = nullptr )
```

Construct a new [List](#) object.

Parameters

<i>destroyValue</i>	If <i>destroyValue</i> != nullptr, the <i>destroyValue</i> function will be called each time a node is removed.
---------------------	---

head is a virtual node that is always present in the list.

Parameters

<i>destroyValue</i>	The destroy function for the values of the list. Defaults to nullptr.
---------------------	---

4.2.2.2 ~List()

```
List::~~List ( )
```

Destroy the [List](#) object.

If the *destroyValue* function is not nullptr, it is called for every value.

4.2.3 Member Function Documentation**4.2.3.1 decreaseSize()**

```
int List::decreaseSize ( )
```

Decrease the list size.

Returns

int The updated size.

4.2.3.2 find()

```
Pointer List::find (
    Pointer value,
    CompareFunc compare )
```

Finds the first value equal to the value parameter.

Find the first value equal to the value parameter.

Parameters

<i>value</i>	The value based on which the search is done.
<i>compare</i>	A pointer to the function that will compare the node values.

Returns

Pointer A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

Pointer A pointer to found value.

4.2.3.3 findNode()

```
ListNode * List::findNode (
    Pointer value,
    CompareFunc compare )
```

Finds the first node that has value equal to the value parameter.

Find the first node with value equal to the value parameter.

Parameters

<i>value</i>	The value to search.
<i>compare</i>	The compare function to use.

Returns

ListNode* A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

ListNode* A pointer to the found node.

4.2.3.4 getHead()

```
ListNode * List::getHead ( )
```

Getter for the head of the [List](#).

Get the first element of the [List](#).

Returns

ListNode* The head.

Returns

ListNode* Pointer to the first element.

4.2.3.5 getSize()

```
int List::getSize ( )
```

Get the size of the [List](#).

Get the size of the list.

The size is updated upon entry/removal of nodes, so the retrieval of the size is O(1).

Returns

int

Returns

int The size.

4.2.3.6 getTail()

```
ListNode * List::getTail ( )
```

Getter for the tail of the [List](#).

Get the last element of the list.

Returns

ListNode* The tail.

If the list is empty, the virtual head is the tail, so we return nullptr.

Returns

ListNode* Pointer to the last element.

4.2.3.7 increaseSize()

```
int List::increaseSize ( )
```

Increase the list size.

Returns

int The updated size.

4.2.3.8 insertNext()

```
void List::insertNext (
    ListNode * node,
    Pointer value )
```

Adds a node *after* the given node.

Insert a new element after the node.

Parameters

<i>node</i>	The node that the new node will be placed after.
<i>value</i>	The value of the new node.

Parameters

<i>node</i>	The node after which the element is inserted.
<i>value</i>	The value of the new element.

4.2.3.9 next()

```
ListNode * List::next (
    ListNode * node )
```

Returns the node after the given one.

Get the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode* A pointer to the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode*

4.2.3.10 nodeValue()

```
Pointer List::nodeValue (
    ListNode * node )
```

Returns the value of the node.

Get the node's value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer A generic pointer to the value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.2.3.11 removeNext()

```
void List::removeNext (
    ListNode * node )
```

Removes the node *after* the given node.

Removes the node after the given node.

Parameters

<i>node</i>	The node before the node to be removed.
-------------	---

Parameters

<i>node</i>	The node after which the element is removed.
-------------	--

4.2.3.12 setDestroyValue()

```
DestroyFunc List::setDestroyValue (
    DestroyFunc value )
```

Set the Destroy Value object to a new value.

Set the destroy value function.

Parameters

<i>value</i>	A pointer to the new destroy function.
--------------	--

Returns

DestroyFunc A pointer to the old destroy function.

Parameters

<i>value</i>	The new destroy function.
--------------	---------------------------

Returns

DestroyFunc The old destroy function.

The documentation for this class was generated from the following files:

- ADTList.hpp
- [ADTList.cpp](#)

4.3 ListNode Class Reference

Node of a list.

```
#include <ADTList.hpp>
```

Public Member Functions

- [ListNode](#) ()
Construct a new [List](#) Node object that is empty.
- [ListNode](#) (Pointer value)
Construct a new [List](#) Node object with value 'value'.
- void [setNext](#) ([ListNode](#) *next)
Setter for the next node.
- [ListNode](#) * [getNext](#) ()
Getter for the next node.
- Pointer [getValue](#) ()
Getter for the value of the node.

4.3.1 Detailed Description

Node of a list.

A list node is represented by its value. It also holds a pointer to the next node in the list.

4.3.2 Constructor & Destructor Documentation

4.3.2.1 [ListNode\(\)](#) [1/2]

```
ListNode::ListNode ( ) [inline]
```

Construct a new [List](#) Node object that is empty.

4.3.2.2 [ListNode\(\)](#) [2/2]

```
ListNode::ListNode (
    Pointer value ) [inline]
```

Construct a new [List](#) Node object with value 'value'.

Parameters

<i>value</i>	a generic pointer to the value of the node.
--------------	---

4.3.3 Member Function Documentation

4.3.3.1 getNext()

```
ListNode * ListNode::getNext ( )
```

Getter for the next node.

Returns

ListNode* The pointer to the next node.

4.3.3.2 getValue()

```
Pointer ListNode::getValue ( )
```

Getter for the value of the node.

Returns

Pointer A generic pointer to the value.

4.3.3.3 setNext()

```
void ListNode::setNext (
    ListNode * next )
```

Setter for the next node.

Parameters

<i>next</i>	The pointer to the next node.
-------------	-------------------------------

The documentation for this class was generated from the following files:

- ADTList.hpp
- [listNode.cpp](#)

4.4 Vector Class Reference

ADT [Vector](#).

```
#include <ADTVector.hpp>
```

Public Member Functions

- [Vector](#) (int size, DestroyFunc destroyValue)
Construct a new [Vector](#) object.
- [~Vector](#) ()
Destroy the [Vector](#) object.
- int [getSize](#) ()
Get the current size of the [Vector](#).
- void [insertLast](#) (Pointer value)
Inserts a new element at the end of the vector.
- int [removeLast](#) ()
Removes the last element of the vector.
- Pointer [getAt](#) (int pos)
Get the value at the specified position of the vector.
- int [setAt](#) (int pos, Pointer value)
Set the value at the specified position of the vector.
- Pointer [find](#) (Pointer value, CompareFunc compare)
Find the first element with value equal to value.
- DestroyFunc [setDestroyValue](#) (DestroyFunc destroyValue)
Set the Destroy Value.
- [vectorNode](#) * [first](#) ()
Get the first node of the vector.
- [vectorNode](#) * [last](#) ()
Get the last node of the vector.
- [vectorNode](#) * [next](#) ([vectorNode](#) *node)
Get the next node of the vector, after the one given.
- [vectorNode](#) * [previous](#) ([vectorNode](#) *node)
Get the previous node of the vector, before the one given.
- Pointer [nodeValue](#) ([vectorNode](#) *node)
Get the value of the node.
- [vectorNode](#) * [findNode](#) (Pointer value, CompareFunc compare)
Find the first node with value equal to value.

4.4.1 Detailed Description

ADT [Vector](#).

An ADT [Vector](#) using a dynamic array for smart resizing.

4.4.2 Constructor & Destructor Documentation

4.4.2.1 [Vector](#)()

```
Vector::Vector (
    int size,
    DestroyFunc destroyValue )
```

Construct a new [Vector](#) object.

The newly created [Vector](#) will be of size size and its elements will be initialized as nullptr.

If destroyValue is not nullptr, it will be called on each element when it is removed from the [Vector](#).

Parameters

<i>size</i>	The initial size of the Vector .
<i>destroyValue</i>	The function to be called on each element when it is removed from the Vector .

4.4.2.2 ~Vector()

```
Vector::~~Vector ( )
```

Destroy the [Vector](#) object.

Deletes all allocated memory of the [Vector](#).

4.4.3 Member Function Documentation**4.4.3.1 find()**

```
Pointer Vector::find (
    Pointer value,
    CompareFunc compare )
```

Find the first element with value equal to value.

Parameters

<i>value</i>	The value to look for.
<i>compare</i>	The function to be used for comparison.

Returns

Pointer A pointer to the found value.

4.4.3.2 findNode()

```
vectorNode * Vector::findNode (
    Pointer value,
    CompareFunc compare )
```

Find the first node with value equal to value.

Parameters

<i>value</i>	The value to look for.
<i>compare</i>	The function to be used for comparison.

Returns

[vectorNode](#) The resulting node.

4.4.3.3 first()

```
vectorNode * Vector::first ( )
```

Get the first node of the vector.

Returns

[vectorNode](#)

4.4.3.4 getAt()

```
Pointer Vector::getAt (
    int pos )
```

Get the value at the specified position of the vector.

Parameters

<i>pos</i>	The position to look for.
------------	---------------------------

Returns

Pointer A pointer to the value at the specified position.

4.4.3.5 getSize()

```
int Vector::getSize ( )
```

Get the current size of the [Vector](#).

Returns

int The size of the [Vector](#).

4.4.3.6 insertLast()

```
void Vector::insertLast (
    Pointer value )
```

Inserts a new element at the end of the vector.

Parameters

<i>value</i>	The value of the new element.
--------------	-------------------------------

If we are at the last element, we create a new array double the size.

4.4.3.7 last()

```
vectorNode * Vector::last ( )
```

Get the last node of the vector.

Returns

[vectorNode](#)

4.4.3.8 next()

```
vectorNode * Vector::next (
    vectorNode * node )
```

Get the next node of the vector, after the one given.

Parameters

<i>node</i>	The node to get the next of.
-------------	------------------------------

Returns

[vectorNode](#)

4.4.3.9 nodeValue()

```
Pointer Vector::nodeValue (
    vectorNode * node )
```

Get the value of the node.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.4.3.10 previous()

```
vectorNode * Vector::previous (
    vectorNode * node )
```

Get the previous node of the vector, before the one given.

Parameters

<i>node</i>	The node to get the previous of.
-------------	----------------------------------

Returns

vectorNode

4.4.3.11 removeLast()

```
int Vector::removeLast ( )
```

Removes the last element of the vector.

Returns

int -1 if the function fails, else 0.

4.4.3.12 setAt()

```
int Vector::setAt (
    int pos,
    Pointer value )
```

Set the value at the specified position of the vector.

Parameters

<i>pos</i>	The position to edit.
<i>value</i>	The new value.

Returns

int -1 if the function fails, else 0.

4.4.3.13 setDestroyValue()

```
DestroyFunc Vector::setDestroyValue (
    DestroyFunc destroyValue )
```

Set the Destroy Value.

Parameters

<i>destroyValue</i>	The new destroy value function.
---------------------	---------------------------------

Returns

DestroyFunc The old destroy value function.

The documentation for this class was generated from the following files:

- [ADTVector.hpp](#)
- [ADTVector.cpp](#)

4.5 vectorNode Class Reference

Class for the vector Node object.

```
#include <ADTVector.hpp>
```

Public Member Functions

- [vectorNode](#) ()
Construct a new vector Node object.
- **vectorNode** (Pointer value)
- Pointer [getValue](#) () const
Get the Value object.
- void [setValue](#) (Pointer value)
Set the Value object.

4.5.1 Detailed Description

Class for the vector Node object.

A vector node is described only its value.

4.5.2 Constructor & Destructor Documentation

4.5.2.1 vectorNode()

```
vectorNode::vectorNode ( ) [inline]
```

Construct a new vector Node object.

Parameters

<i>value</i>	A Pointer to the value.
--------------	-------------------------

4.5.3 Member Function Documentation

4.5.3.1 getValue()

```
Pointer vectorNode::getValue ( ) const [inline]
```

Get the Value object.

Returns

Pointer

4.5.3.2 setValue()

```
void vectorNode::setValue (
    Pointer value ) [inline]
```

Set the Value object.

Parameters

<i>value</i>	
--------------	--

The documentation for this class was generated from the following file:

- [ADTVector.hpp](#)

Chapter 5

File Documentation

5.1 ADTList.hpp

```
00001
00011 #pragma once
00012 #include "common.hpp"
00013
00014 #define LIST_BOF (ListNode*)0
00015 #define LIST_EOF (ListNode*)0
00016
00024 class ListNode {
00025 public:
00030     ListNode() : next(nullptr), value(nullptr){};
00036     ListNode(Pointer value) : next(nullptr), value(value){};
00042     void setNext(ListNode* next);
00048     ListNode* getNext();
00054     Pointer getValue();
00055
00056 private:
00057     ListNode* next;
00058     Pointer value;
00059 };
00060
00068 class List {
00069 public:
00076     List(DestroyFunc value = nullptr);
00081     ~List();
00088     void insertNext(ListNode* node, Pointer value);
00094     void removeNext(ListNode* node);
00102     Pointer find(Pointer value, CompareFunc compare);
00109     DestroyFunc setDestroyValue(DestroyFunc value);
00118     int getSize();
00119     int increaseSize();
00120     int decreaseSize();
00126     ListNode* getHead();
00132     ListNode* getTail();
00139     ListNode* next(ListNode* node);
00146     Pointer nodeValue(ListNode* node);
00154     ListNode* findNode(Pointer value, CompareFunc compare);
00155
00156 private:
00157     DestroyFunc destroyValue;
00158     int size;
00159     ListNode* head;
00160     ListNode* tail;
00161 };
```

5.2 ADTVector.hpp File Reference

ADTVector implementation using a dynamic array.

```
#include "common.hpp"
```

Include dependency graph for ADTVector.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [vectorNode](#)
Class for the vector Node object.
- class [Vector](#)
ADT [Vector](#).

Macros

- `#define VECTOR_MIN_CAPACITY 10`
- `#define VECTOR_BOF (vectorNode\*)0`
- `#define VECTOR_EOF (vectorNode\*)0`

5.2.1 Detailed Description

ADTVector implementation using a dynamic array.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

Copyright

Copyright (c) 2023

5.3 ADTVector.hpp

[Go to the documentation of this file.](#)

```
00001
00011 #pragma once
00012
00013 #include "common.hpp"
00014
00015 #define VECTOR_MIN_CAPACITY 10
00016
00017 #define VECTOR_BOF (vectorNode\*)0
00018 #define VECTOR_EOF (vectorNode\*)0
00019
00026 class vectorNode {
00027 public:
00033     vectorNode(){};
00034     vectorNode(Pointer value) : value(value){};
00040     Pointer getValue() const { return value; };
00046     void setValue(Pointer value) { this->value = value; };
00047
00048 private:
00049     Pointer value;
00050 };
00051
```



```

00058 class Vector {
00059 public:
00073     Vector(int size, DestroyFunc destroyValue);
00080     ~Vector();
00086     int getSize();
00092     void insertLast(Pointer value);
00098     int removeLast();
00105     Pointer getAt(int pos);
00113     int setAt(int pos, Pointer value);
00121     Pointer find(Pointer value, CompareFunc compare);
00128     DestroyFunc setDestroyValue(DestroyFunc destroyValue);
00134     vectorNode* first();
00140     vectorNode* last();
00147     vectorNode* next(vectorNode* node);
00154     vectorNode* previous(vectorNode* node);
00161     Pointer nodeValue(vectorNode* node);
00169     vectorNode* findNode(Pointer value, CompareFunc compare);
00170
00171 private:
00172     vectorNode* array;
00173     int size;
00174     int capacity;
00175     DestroyFunc destroyValue;
00176 };

```

5.4 common.hpp

```

00001 #pragma once
00002
00003 typedef void* Pointer;
00004
00005 typedef int (*CompareFunc)(Pointer a, Pointer b);
00006
00007 typedef void (*DestroyFunc)(Pointer value);

```

5.5 cppdescent.hpp

```

00001
00006 class cppdescent {
00007 public:
00013     int hello(void);
00014 };

```

5.6 ADTList.cpp File Reference

Implementation of an ADT linked list.

```
#include "cppdescent/ADTList.hpp"
```

Include dependency graph for ADTList.cpp:

5.7 ADTVector.cpp File Reference

Implementation of ADTVector using a dynamic array.

```
#include "cppdescent/ADTVector.hpp"
#include "stdlib.h"
```

Include dependency graph for ADTVector.cpp:

5.7.1 Detailed Description

Implementation of ADTVector using a dynamic array.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-30

Copyright

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5.8 cppdescent.cpp File Reference

Implementation of cppdescent library.

```
#include "cppdescent/cppdescent.hpp"  
#include <iostream>  
Include dependency graph for cppdescent.cpp:
```

5.8.1 Detailed Description

Implementation of cppdescent library.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-27

Copyright

Copyright (c) 2023

5.9 listNode.cpp File Reference

Implementation of the listNode to be used with linkedList.

```
#include "cppdescent/ADTList.hpp"
```

Include dependency graph for listNode.cpp:

5.9.1 Detailed Description

Implementation of the listNode to be used with linkedList.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

Copyright

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